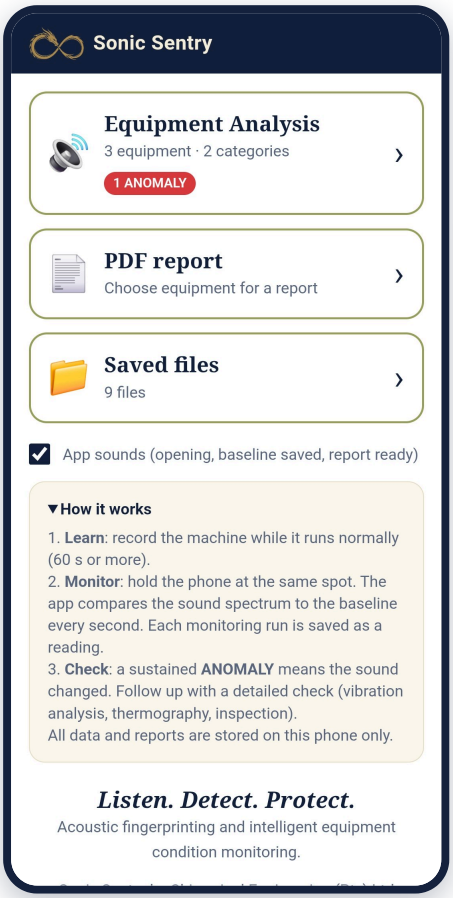
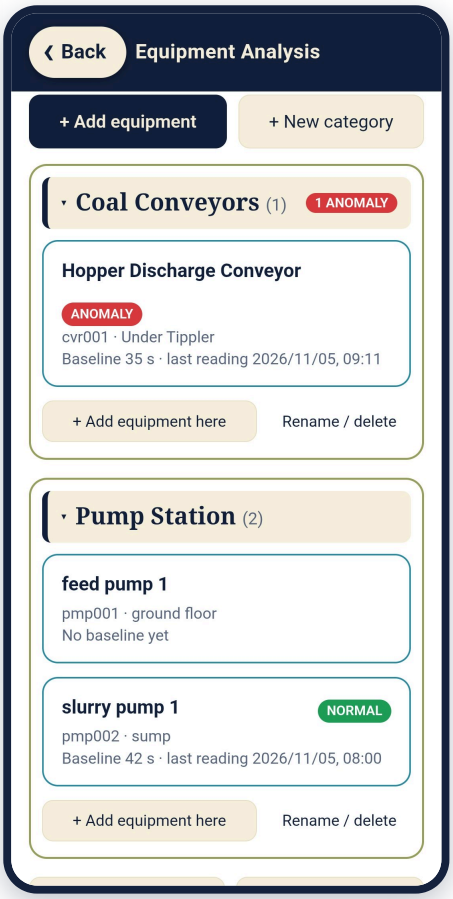
Machines often sound different before they fail.

Sonic Sentry turns an Android phone into an acoustic screening tool. It learns how each machine sounds when healthy, then shows you when that sound changes.

Walk-down rounds depend on experienced ears, and that experience is hard to pass between shifts. Sonic Sentry gives every technician the same reference: a saved sound signature for each machine.







Starts with a phone

No sensors to fit and no cabling. Use the phone microphone, or plug in an external microphone.

Shows what changed

See which frequency range moved, by how much, and what that range usually points to.

Keeps data with you

No accounts and no internet needed. No audio is recorded, and nothing leaves the phone unless you share it.

Three steps from sound to decision

The app measures 60 frequency bands from 20 Hz to 20 kHz and compares them with the machine's baseline every second.

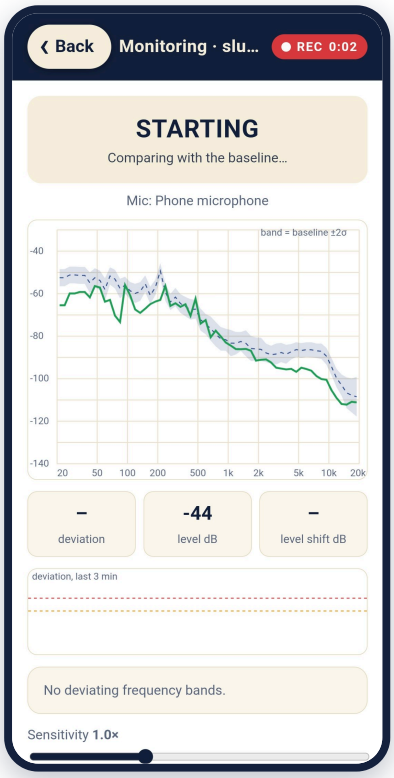
1 Learn

Hold the phone at a fixed spot for 60 seconds or more while the machine runs normally. Its sound spectrum becomes the baseline.



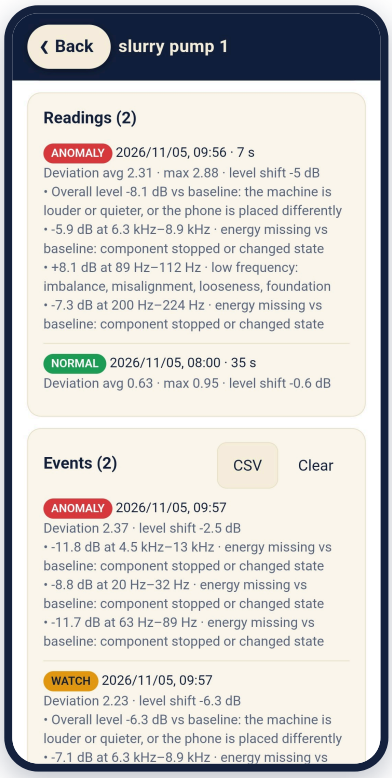
2 Monitor

Return to the same spot on your next round. The live spectrum is drawn over the normal band, so a change is easy to see.



3 Check

Every run is saved as a reading. Findings list the changed ranges, so you know where to aim a detailed check.



Clear status

- **NORMAL** Sound matches the baseline.
- **WATCH** A sustained change. Keep an eye on it.
- **ANOMALY** A larger sustained change. Plan a check.

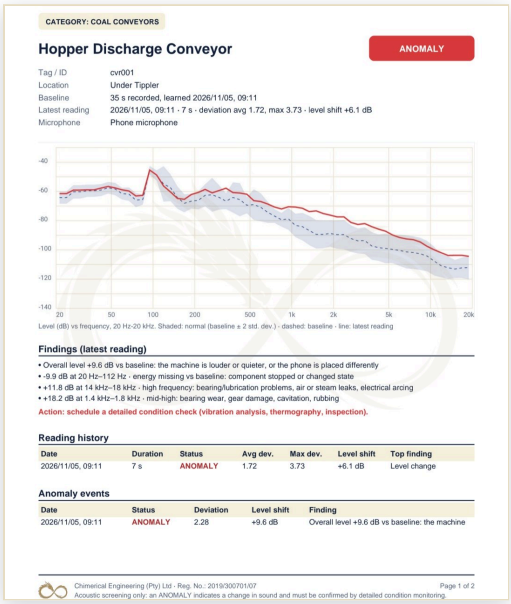
A change must last 3 seconds, so a single bang does not raise a flag.

What a louder range usually points to

Below 50 Hz	Slow-speed imbalance, structural or foundation movement, pulsation
50–200 Hz	Imbalance, misalignment, looseness, foundation
200 Hz–1 kHz	Gear mesh, belts, flow or pump noise, structural resonance
1–4 kHz	Bearing wear, gear damage, cavitation, rubbing
Above 4 kHz	Bearing or lubrication problems, air or steam leaks, electrical arcing

Reports your team can act on

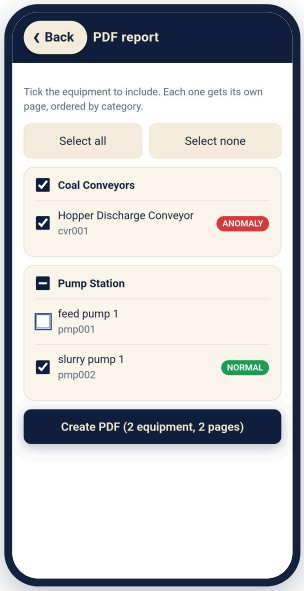
Tick the equipment you want and create a branded PDF on the phone, with one page per machine.



An anomaly page: spectrum against baseline, findings and a recommended action.



A normal page: a record that the machine matched its baseline.



Built for the plant floor

- Group equipment into your own categories, such as areas or systems
- Status at a glance for every machine and category
- Reading history and an anomaly event log for each machine
- Adjustable sensitivity for quiet or noisy areas
- Silent during monitoring: no sound or vibration to affect the reading
- Saved files sorted by date and filtered by category
- Share by email, WhatsApp or Drive; export events as CSV

Who it is for

Maintenance teams, reliability engineers and plant technicians looking after pumps, conveyors, fans, motors and gearboxes.

A screening tool, not a diagnosis. An ANOMALY means the sound changed and a detailed check is needed (vibration analysis, thermography, inspection). Use the same phone and the same spot each time. Phone microphones are weaker at very low frequencies.

CHIMERICAL ENGINEERING (PTY) LTD
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See it on your own equipment

Ask us for a demonstration of Sonic Sentry.

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